

8th Grade Physical Science — Vocabulary Reference

Georgia Standards of Excellence (S8P1 – S8P5) • Five Unit Vocabulary

Name: _____

Period: _____

Date: _____

Section 1 — Matter & the Periodic Table

S8P1

Matter: anything that has mass and takes up space (volume).

Atom: the smallest unit of matter that keeps the properties of an element.

Element: a pure substance made of only one type of atom; cannot be broken down by chemical means.

Compound: a pure substance made of two or more elements chemically bonded together (e.g., H₂O, CO₂).

Pure Substance: matter with a fixed composition — either an element or a compound.

Mixture: two or more substances physically combined; can be separated by physical means.

Homogeneous Mixture: a mixture that looks the same throughout (e.g., saltwater); also called a solution.

Heterogeneous Mixture: a mixture where different parts can be seen (e.g., trail mix, sand and water).

Solution: a homogeneous mixture of a solute dissolved in a solvent.

Proton: a positively charged particle in the nucleus of an atom.

Neutron: a neutral (no charge) particle in the nucleus of an atom.

Electron: a negatively charged particle that moves around the nucleus.

Nucleus: the dense center of the atom containing protons and neutrons.

Atomic Number: the number of protons in an atom; identifies the element.

Atomic Mass: the total mass of an atom (protons + neutrons).

Periodic Table: an organized chart of all the known elements arranged by atomic number.

Group (Family): a vertical column on the periodic table; elements have similar properties.

Period: a horizontal row on the periodic table.

Metal: shiny, conducts heat and electricity, malleable; found on the left of the periodic table.

Nonmetal: dull, poor conductor, brittle; found on the right side of the periodic table.

Metalloid: element with both metal and nonmetal properties (along the staircase).

Molecule: two or more atoms chemically bonded together.

Solid: state of matter with definite shape and volume; particles vibrate in place.

Liquid: state of matter with definite volume but no definite shape; particles slide past each other.

Gas: state of matter with no definite shape or volume; particles move freely.

Plasma: superheated gas with charged particles; found in stars and lightning.

Physical Property: a characteristic that can be observed without changing the substance (color, density, melting point, boiling point).

Chemical Property: a characteristic that describes how a substance reacts with other substances (reactivity, combustibility).

Physical Change: a change in form or appearance only; no new substance is made (melting, cutting, dissolving).

Chemical Change: a change that produces a new substance (rusting, burning, digestion).

Density: amount of mass in a given volume; $D = m / V$.

Melting Point: temperature at which a solid becomes a liquid.

Boiling Point: temperature at which a liquid becomes a gas.

Reactivity: how easily a substance reacts chemically with other substances.

Combustibility: the ability of a substance to burn.

Chemical Reaction: a process where reactants are changed into new products.

Reactant: a substance that goes into a chemical reaction (left side of equation).

Product: a substance produced by a chemical reaction (right side of equation).

Law of Conservation of Mass: matter is not created or destroyed in a chemical reaction; mass of reactants = mass of products.

Chemical Equation: a representation of a chemical reaction using symbols and formulas.

Section 2 — Energy

S8P2

Energy: the ability to do work or cause change.

Work: force applied to an object that causes it to move in the direction of the force.

Kinetic Energy (KE): the energy of motion; depends on mass and speed ($KE = \frac{1}{2}mv^2$).

Potential Energy (PE): stored energy due to position or condition.

Gravitational Potential Energy: stored energy of an object due to its height above the ground; depends on mass and height.

Mechanical Energy: the total of an object's kinetic and potential energy.

Law of Conservation of Energy: energy cannot be created or destroyed; it can only change forms.

Energy Transformation: a change from one form of energy to another (e.g., chemical → light).

Thermal Energy: total energy of the particles in a substance; related to motion of atoms.

Heat: thermal energy that transfers from a warmer object to a cooler object.

Temperature: the average kinetic energy of the particles in a substance.

Conduction: transfer of heat through direct contact between particles (collision of atoms).

Convection: transfer of heat through currents in a fluid (liquid or gas).

Radiation: transfer of heat through electromagnetic waves; can travel through empty space.

Chemical Energy: energy stored in the bonds of atoms and molecules (food, fuel, batteries).

Electrical Energy: energy carried by moving electric charges.

Light Energy (Radiant): energy carried by electromagnetic waves we can see.

Sound Energy: energy carried by vibrations through a medium.

Nuclear Energy: energy stored in the nucleus of an atom.

Elastic Potential Energy: stored energy in an object that can be stretched or compressed (rubber bands, springs).

System: a group of related objects or parts that work together.

Joule (J): the SI unit for measuring energy and work.

Mass: the amount of matter in an object (affects both KE and PE).

Friction: a force that converts kinetic energy into thermal energy.

Section 3 — Force, Mass & Motion

S8P3

Motion: a change in an object's position over time relative to a reference point.

Reference Point: a fixed place or object used to determine if motion has occurred.

Position: the location of an object.

Distance: the total length of the path traveled by an object.

Displacement: the change in position of an object from start to finish (has direction).

Speed: how fast an object moves; distance divided by time ($s = d/t$).

Average Speed: total distance divided by total time.

Velocity: speed in a specific direction.

Acceleration: the rate at which velocity changes over time (speeding up, slowing down, or changing direction).

Deceleration: a decrease in speed; negative acceleration.

Force: a push or pull that can cause an object to move, stop, or change direction.

Newton (N): the SI unit used to measure force.

Net Force: the overall force acting on an object after all forces are combined.

Balanced Forces: forces that are equal and opposite; net force is zero; no change in motion.

Unbalanced Forces: forces that are not equal; cause an object to change its motion.

Friction: a force that opposes motion between two surfaces in contact.

Inertia: the tendency of an object to resist changes in its motion; depends on mass.

Mass: the amount of matter in an object; measured in kilograms (kg).

Weight: the force of gravity acting on an object's mass.

Newton's First Law: an object at rest stays at rest, and an object in motion stays in motion, unless acted on by an unbalanced force (Law of Inertia).

Newton's Second Law: force equals mass times acceleration ($F = ma$); more mass needs more force to accelerate.

Newton's Third Law: for every action there is an equal and opposite reaction.

Action Force: the initial force applied by one object on another.

Reaction Force: the equal and opposite force applied back by the second object.

Gravity: a force that pulls objects with mass toward one another.

Air Resistance: a type of friction caused by air pushing against a moving object.

Free Fall: motion of an object falling under the influence of gravity only.

Section 4 — Waves (Light & Sound)

S8P4

Wave: a disturbance that transfers energy from one place to another without transferring matter.

Medium: the material through which a wave travels (solid, liquid, or gas).

Mechanical Wave: a wave that needs a medium to travel through (sound, water waves).

Electromagnetic Wave: a wave that does not need a medium; can travel through empty space (light, radio).

Transverse Wave: a wave where particles move perpendicular (up/down) to the direction of the wave (light, water).

Longitudinal Wave: a wave where particles move parallel to the direction of the wave (sound).

Crest: the highest point of a transverse wave.

Trough: the lowest point of a transverse wave.

Compression: the part of a longitudinal wave where particles are pushed together.

Rarefaction: the part of a longitudinal wave where particles are spread apart.

Wavelength: the distance from one crest to the next (or compression to compression).

Frequency: the number of waves that pass a point in one second; measured in Hertz (Hz).

Amplitude: the height of a wave from its rest position; relates to energy or loudness/brightness.

Wave Speed: how fast a wave travels; depends on the medium.

Electromagnetic Spectrum: the full range of EM waves, from radio waves to gamma rays.

Radio Waves: EM waves with the longest wavelength and lowest energy; used for communication.

Microwaves: EM waves used for cooking and cell phones.

Infrared: EM waves we feel as heat.

Visible Light: the only part of the EM spectrum humans can see (ROYGBIV).

Ultraviolet (UV): EM waves with more energy than visible light; can cause sunburn.

X-rays: high-energy EM waves used in medical imaging.

Gamma Rays: EM waves with the shortest wavelength and highest energy.

Reflection: the bouncing of a wave off a surface.

Refraction: the bending of a wave as it passes from one medium to another.

Absorption: when a wave's energy is taken in by a material.

Diffraction: the bending of a wave around an obstacle or through an opening.

Transmission: when a wave passes through a material.

Pitch: how high or low a sound is; depends on the frequency of the sound wave.

Volume (Loudness): how loud or soft a sound is; depends on amplitude.

Echo: a reflected sound wave.

Lens: a curved piece of clear material that refracts light to form an image.

Convex Lens: a lens that is thicker in the middle; converges light (magnifying glass, eye).

Concave Lens: a lens that is thinner in the middle; diverges light (corrects nearsightedness).

Image: the picture formed when light rays come together through a lens or mirror.

Opaque: a material that does not let light pass through.

Translucent: a material that lets some light through but scatters it (frosted glass).

Transparent: a material that lets light pass through clearly (window glass).

Vacuum: an empty space with no matter; EM waves can travel through, mechanical waves cannot.

Section 5 — Forces in Nature (Gravity, Electricity & Magnetism)

S8P5

Field: an area around an object where its force can affect other objects without touching them.

Non-contact Force: a force that acts on objects without touching them (gravity, magnetism, electric force).

Gravity: a force of attraction between any two objects that have mass.

Gravitational Field: the region around an object with mass where another object will feel a gravitational pull.

Weight: a measure of the force of gravity on an object; changes with location.

Mass: the amount of matter in an object; stays the same everywhere.

Electric Charge: a property of matter that causes it to feel a force in an electric field; can be positive (+) or negative (–).

Positive Charge: the charge carried by protons.

Negative Charge: the charge carried by electrons.

Neutral: having an equal number of positive and negative charges.

Electric Force: the attractive or repulsive force between charged objects.

Electric Field: the area around a charged object where its electric force can be felt.

Static Electricity: the buildup of electric charges on the surface of an object.

Conductor: a material through which electric charges flow easily (most metals, copper, silver).

Insulator: a material that does not allow electric charges to flow easily (rubber, plastic, wood).

Electric Current: the flow of electric charges through a conductor.

Circuit: a complete path through which electric current can flow.

Voltage: the push or energy that moves electric charges through a circuit.

Magnet: an object that produces a magnetic field and attracts iron or steel.

Magnetic Field: the area around a magnet where its magnetic force can be felt.

Magnetic Pole: one of two ends of a magnet (north and south) where the magnetic force is strongest.

Attraction: a force that pulls objects together (opposite charges / opposite poles).

Repulsion: a force that pushes objects apart (like charges / like poles).

Electromagnet: a temporary magnet created when an electric current flows through a coiled wire, often around an iron core.

Solenoid: a coil of wire that produces a magnetic field when current flows through it.

Electromagnetism: the relationship between electricity and magnetism; moving charges create magnetic fields.

Dry Cell: a battery that stores chemical energy and converts it to electrical energy.

Iron Core: a piece of iron placed inside a coil of wire to make an electromagnet stronger.

Distance: a factor that affects the strength of gravity, electric, and magnetic forces — as distance increases, force decreases.